$$\boxed{a_n = \boxed{}\, a_{n-1} + \boxed{}\, a_{n-2}} \quad n \geq 3.$$

as : $a_1 = 2$, $a_2 = 3$ } First two values of right most column in prev. table. Only two bcs we've two terms in a_n.

$\longleftrightarrow$

Ex: 8.2 "SOLVING LINEAR NON HOMO. & HOMOGENEOUS R.R WITH CONSTANT COFFECIENT".

DEF: A linear homo. recurrence relation of degree 'k' with constant cofficient is a RR of the form:

$$\boxed{a_n = C_1 a_{n-1} + C_2 a_{n-2} + \ldots\ldots C_k a_{n-k}}$$

where $C_1, C_2 \ldots\ldots C_k \in \mathbb{R}$ and $C_k \neq 0$.

Example: $f_n = f_{n-1} + f_{n-2}$ (LHCC)

$a_n = a_{n-1} + a_{n-2}^2$ (Non Linear) as highest power '2'

$b_n = n b_{n-1}$ (Non Linear cofficients are func. of 'n').

DEF: A linear Non-Homo R.R of degree 'k' with constant coff. is a R.R of the form:

$$\boxed{a_n = C_1 a_{n-1} + C_2 a_{n-2} + \ldots\ldots + C_k a_{n-k} + F(n)}$$

where $C_1, C_2, \ldots\ldots C_k \in \mathbb{R}$ and $F(n)$ is a func. not identically zero depending only on 'n'.

Example: $a_n = a_{n-1} + 2n$ (L. Non-Homo)

$a_n = 5a_{n-1} - 6a_{n-2} + 7^n$ → these func. in eq made them non-homo.

⟹ Theorum 3 : (Unique roots)

Let $C_1, C_2 \ldots C_k$ be real no. Suppose the characteristics poly/eq. is $r^k - C_1 r^{k-1} - C_2 r^{k-2} \ldots (k=0)$ has 'k' distinct roots $r_1, r_2, r_3 \ldots\ldots r_k$. Then a seq $\{a_n\}$ is a sol. of RR:

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_k a_{n-k}$ iff

Ⓐ $\boxed{a_n = L_1 r_1^n + L_2 r_2^n + \ldots\ldots L_k r_k^n}$ for

$n = 0, 1, 2 \ldots$ where $L_1, L_2, L_3 \ldots L_k$ constants.

→ Theorum 1 : Suppose $C_1, C_2 \in \mathbb{R}$. Suppose the char. ~~[illegible]~~ poly is $r^2 - C_1 r - C_2 = 0$. has 2 distinct roots r_1 & r_2.

Then a seq $\{a_n\}$ is a sol of RR:

$\boxed{a_n = C_1 a_{n-1} + C_2 a_{n-2}}$ iff $a_n = L_1 r_1^n + L_2 r_2^n$ for $n = 0, 1, 2, \ldots$ where L_1, L_2 are constants.

→ Theorem 2 : Suppose $c_1, c_2 \in \mathbb{R}$. Suppose the char. poly is $r^2 - c_1 r - c_2 = 0$ has repeated roots. The a seq. $\{a_n\}$ is a sol of R.R: → Repeated roots.

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

iff

$$\boxed{a_n = L_1 r_1^n + L_2 n r_1^n}$$

for $n = 0, 1, 2, \dots$

where L_1 & L_2 are constants

METHOD :

① Write characteristics polynomial (put poly. set equal to zero)

② Factorize polynomial & find roots (by any method!). OR calculator.

③ Determine form of a_n:
→ Distinct → Repeated → Complex.

④ Solve coefficients (if initial conditions given)

⑤ Substitute values in a_n.

Q: What is the sol. of R.R $a_n = a_{n-1} + 2a_{n-2}$ with $a_0 = 2$, $a_1 = 7$. degree = 2

Take subscript to the power of 'r'

→ degree n=2.

$$r^n = r^{n-1} - 2r^{n-2} = 0.$$

$$r^2 = r^{2-1} - 2r^{2-2} = 0$$

$$\boxed{r^2 - r - 2 = 0}$$ → characteristic Polynomial.

$$r^2 - 2r + r - 2 = 0$$

$$r(r-2) + 1(r-2) = 0.$$

$\boxed{r = -1}$, $\boxed{r = 2}$

★ Eq (A) from Theorem 3. (as distinct roots).

$$a_n = L_1 r_1^n + L_2 r_2^n$$

• put roots in "r_1" and "r_2"

$a_n = L_1 (-1)^n + L_2 (2)^n$ → would be our final answer if initial conditions a_0 & a_1 were NOT given in ques.

⊙ set n=0

$a_0 = L_1(-1)^0 + L_2(2)^0$

$2 = L_1 + L_2$ ⟶ (A)

⊙ Set n = 1

$a_1 = L_1(-1)^1 + L_2(2)^1$

$7 = -L_1 + 2L_2$ ⟶ (B)

★ Solving (A) & (B):

$L_1 = -1$, $L_2 = 3$

$\therefore$ $\boxed{a_n = -(-1)^n + 3(2^n)}$ FINAL ANSWER.

Q: What is Sol. of RR : $6a_{n-1} - 9a_{n-2}$
with I.C $a_0 = 1$, $a_1 = 6$? degree $n=2$

$\boxed{r^2 - 6r + 9 = 0}$ Char. Poly

Roots: $(r-3)^2 = 0$
$\boxed{r = 3}$ $\boxed{r=3}$ * as repeated roots from Theorem 2 formula.

$a_n = L_1 (3)^n + L_2 n (3)^n$

→ Solving with IC:

- Set $n=0$
  $a_0 = L_1 (3)^0 + L_2 (0)(3)^0$
  $1 = L_1$ ⟶ (A)

- Set $n=1$
  $a_1 = L_1 (3)^1 + L_2 (1)(3)^1$
  $6 = 3L_1 + 3L_2$ ⟶ (B)

* Solving (A) & B

$\boxed{L_1 = 1}$, $\boxed{L_2 = 1}$.

NOTE: for general form of a complex homo. eq; we'll equate it to zero (characteristic eq) & put values of r in $a_n = \alpha (r_1)^n + \beta (r_2)^n$ .... find this will be our general eq.

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Q(1): Find the Sol. to : $a_n = 5a_{n-2} + 4a_{n-4}$.
with $a_0 = 3$, $a_1 = 2$, $a_2 = 6$, $a_3 = 8$

⟹ Theorem 4 : Distinct & Simple roots Both.

Q: Characteristic Poly. of Linear Homo. has roots: 2, 2, 2, 5, 5, 9.

General Sol:
$[L_{1,0} + L_{1,1}(n) + L_{1,2}(n^2)]\, 2^n +$ first root
$[L_{2,0} + L_{2,1}(n)]\, 5^n +$ second root
$[L_{3,0}]\, 9^n$. third root

LECTURE-22

Wednesday (12/11/25)

☐ Non-Homogeneous Linear R.R with Constant Coff.: of form:

$a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots C_k a_{n-k} + F(n)$

Consists of homogeneous part and associative part:

$a_n^{(h)} = C_1 a_{n-1} + C_2 a_{n-2} + \cdots C_k a_{n-k}$ is homogeneous
$a_n^{(p)} = F(n)$

$\boxed{\text{Step 1}}$: Solve homo. part a_n^{h}

$\boxed{\text{Step 2}}$: Take suitable form for $F(n)$

	$F(n)$	$a_n^{(p)}$
constant	C	A
linear	n / $(2n+1)$	$An+B$
quadratic	n^2	An^2+Bn+C
	r^n	$C\,r^n$

$\boxed{\text{Step 3}}$: Solve $a_n^{(p)}$

$\boxed{\text{Step 4}}$: Substitute $a_n = a_n^{(h)} + a_n^{(p)}$

$\boxed{\text{Step 5}}$: Solve Initial Condition (if given).

Q: Find all Solutions of R.R

$$a_n = 5a_{n-1} \mp 6a_{n-2} + 7^n.$$

$a_n^{(h)}$: $a_n = 5a_{n-1} + 6a_{n-2}$ solving homo part ①

$r^2 - 5r + 6 = 0$

$r^2(r-3) - 2(r-3) = 0$

$\boxed{r=2}$ $\boxed{r=3}$

$a_n^{(h)} = \alpha r_1^n + \beta r_2^n$

$\boxed{a_n^{(h)} = \alpha 2^n + \beta 3^n}$ → don't solve for α & β yet. do it at end.

$F(n) = 7^n$ ② suitable $F(n)$ form

$a_n^{(p)} = C7^n$

$C7^n = 5C7^{n-1} - 6C7^{n-2} + 7^n.$ (from ques) → Ⓐ

★ Divide Ⓐ by 7^{n-2} ③ Solving $a_n^{(p)}$

should be 7^{n} – highest no.

$$\frac{C7^n}{7^{n-2}} = \frac{5C7^{n-1}}{7^{n-2}} - \frac{6C7^{n-2}}{7^{n-2}} + \frac{7^n}{7^{n-2}}$$

$$C7^{n-(n-2)} - 5C7^{(n-1)-(n-2)} + 6C7^{(n-2)-(n-2)} = 7^{n-(n-2)}$$

$7^2C - 5C(7) + 6C = 7^2$ (Take const. to one side)

$49C - 35C + 6C = 49$

$$\boxed{C = \frac{49}{20}}$$

④ Substitute 'C' in both.

$a_n = a_n^{(h)} + a_n^{(p)}$

answer: $a_n = \left[\alpha 2^n + \beta 3^n\right] + \left[\frac{49}{20} 7^n\right]$

⑤ if initial conditions were given, then now we w'ld've solved for them

Q: Find all Sols. of R.R: ; $a_1 = 3$ $n \geq 1$

$$a_n = 3a_{n-1} + 2n$$

① Solving Homo part

$r - 3 = 0$

$\boxed{r=3}$

$\therefore \boxed{a_n^{(h)} = \alpha 3^n}$

② Suitable $F(n)$: $An+B$ as linear (i.e $2n$)

replace 'n' in ques. with 'n-1'.

$$An+B = 3(A(n-1)+B) + 2n$$

③ Solve $a_n^{(p)}$

$$(An - 3An - 2n) + (B + 3A - 3B) = 0.$$

$$(-2A-2)n + (3A-2B) = 0.$$

$$-2(A+1)n + (3A-2B) = 0$$

Comparing :

$A+1=0$; $\boxed{A=-1}$

$3A-2B=0$.

$3(-1) = 2B$

$\boxed{B = \frac{-3}{2}}$ $\therefore$ $\boxed{a_n^{(p)} = -n - \frac{3}{2}}$

④ Combining both $a_n^{(p)}$ & $a_n^{(h)}$

$$a_n = [\alpha 3^n] + \left[-n - \frac{3}{2}\right]$$

⑤ Initial Conditions.

$a_1 = 3$; to find α.

$$a_1 = \alpha 3^1 - 1 - \frac{3}{2}$$

$$3 = 3\alpha - \frac{5}{2}$$

$3\alpha = 3 + 5/2$

$3\alpha = 11/2$

$\boxed{\alpha = 11/6}$.

Sol: $a_n = \frac{11}{6}(3^n) - n - \frac{3}{2}$

CHP#9: "Relations"

Defination: Let A and B be sets. A binary relation from A ~~to~~ to B is a subset of $A \times B$

- A binary relation from A to B is a set of R of ordered pairs:

Example. $A = \{0, 1, 2\}$ $B = \{a, b\}$.

cartesian product. $A \times B = \{(0,a), (0,b), (1,a), (1,b), (2,a), (2,b)\}$.

and Let $R = \{(0,a), (0,b), (1,a), (1,b)\}$

subset

$$R \subseteq A \times B$$

we say $0Ra$ if $(0,a) \in R$

$1 \not R b$ if $(1,b) \notin R$

Relation on a Set (itself): A relation on a set A is a relation from A to A.

Example: Let $A = \{1, 2, 3, 4\}$.

which ordered pairs are in the relation.

$R_1 = \{(a,b) \mid a \text{ divides } b\}$

$R_2 = \{(a,b) \mid a > b\}$.

$R_3 = \{(a,b) \mid a = b\}$

as 1|1 as 2|4

$R_1 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

$R_2 = \{(2,1), (3,1), (4,1), (3,2), (4,2), (4,3)\}$

$R_3 = \{(1,1), (2,2), (3,3), (4,4)\}$.

↳ set like these holds all properties.

• Transitive closure of set R: Make a set which includes elements that satisfies transitive property on existing set 'R'.

• Equivalence relation: If set is reflexive, symmetric, transitive

Ex: 9.1

LECTURE - 23
Monday 17/11/25.

Relation Properties: (on Sets)

1) Reflexive: A relation R on a set A is called reflexive if $(a,a) \in R$ for every element $a \in A$.
Jitney bhi A kay elements; unka double similar (a,a) shld exist in R.

2) Symmetric: A relation R on a set A is symmetric if $(b,a) \in R$ whenever $(a,b) \in R$, for all $a,b \in A$. If $(a,b) \in R$ then $(b,a) \in R \rightarrow$ e.g.: (1,2) & (2,1) both should $\in R$.

3) Antisymmetric: A relation R on a set A is antisymmetric such that for all $a,b \in A$; if $(a,b) \in R$ and $(b,a) \in R$ then $a=b$. Implication logic table.
$[(a,b) \in R \wedge (b,a) \in R] \longrightarrow a=b$.

4) Transitive: A relation R on set A is called transistive if whenever $(a,b) \in R$ and $(b,c) \in R$, then $(a,c) \in R$, for all $a,b,c \in R$.
$[(a,b) \in R \wedge (b,c) \in R] \longrightarrow (a,c) \in R$.

→ Relations of Sets (from prev. page): [R1 example]

Reflexive
• In R_1; since (1,1), (2,2), (3,3), (4,4) all exists in R_1 ∴ R_1 set is reflexive. (Since $\forall a \in A, (a,a) \in R$)
if we wldn't have either ordered pair like (2,2) in R_1; R_1 wldn't be reflexive.

If not symmetric; then definitely antisymmetric
If symmetric; then anti-symmetric if $a=b$.

Symmetric
• Any one example needed to prove R_1 is not symmetric. Since $(1,2) \in R_1$ but $(2,1) \notin R_1$ ∴ R_1 is not symmetric.

Anti-Symmetric

$(a,b) \in R$	$(b,a) \in R$	Is antisymetric.
True	True	True
True	False	False
False	True	True
False	False	True

$(1,2) \in R_1$ ∴ $(a,b) \in R_1$ ✓
$(2,1) \notin R_1$ ∴ $(b,a) \notin R_1$ X
} answer of 'n' = false

$1 \neq 2$ ∴ $a \neq b$ } false.

false → false = True ∴ It is antisymetric

Transitive
• $(1,2) \wedge (2,4)$ = True as both belongs to R_1.
after transitive ($a,c \in R_1$) ; (2,4) ∴ it is also
True . True → True = True ∴ Transitive

$(1,3) \wedge (3,3) \rightarrow (3,3)$
True True = True.

Note: no need to check pairs like (1,1) (2,2), ...-a=b in transitive

Check all elements of ordered pairs in a set.

Q: $R = \{(2,1), (3,1), (4,1), (3,2), (4,2), (4,3)\}$.

① Non-Reflexive as $(1,1), (2,2), (3,3), (4,4) \notin R$

② Symmetric (NOT) as $(2,1) \in R$ but $(1,2) \notin R$.

③ Antisymmetric: $(2,1) \wedge (1,2) \in R \rightarrow (1 = 2)$

$T \quad F \longrightarrow F$

$F \rightarrow F$ = True

④ Transitive as $(1,1) \wedge (1,1) \rightarrow (1,1) \in R$.

$T \quad T \rightarrow T$ = True.

Q: $A = \{1, 2, 3, 4\}$

$R_1 = \{(a,b) \mid a = b + 1\}$

$R_2 = \{(a,b) \mid a+b \leq 3\}$.

$R_1 = \{(2,1), (3,2), (4,3)\}$

$R_2 = \{(1,1), (1,2), (2,1)\}$

$\boxed{R_2}$ ∵ • Not reflexive as similar pairs $\notin R_2$

• Symmetric as $(1,2) \in R_2$ & $(2,1) \in R_2$

• Anti-Symmetric (NOT) as $(2,1) \wedge (1,2) \rightarrow 1=2)$

$T \wedge T \rightarrow F$ = F

and rest examples are also trues with same logic ✗

• Transitive (NOT) as $(2,1) \wedge (1,2) \rightarrow (2,2) \notin R$

$T \wedge T \longrightarrow F$

$T \rightarrow F$ = false

9.2 not in Syllabus-

9.3 :

"Representation Relations:

A relation b/w finite sets can be represented using a zero-one matrix.

Let R is a relation from $A = \{a_1, a_2 \dots a_m\}$ to $B = \{b_1, b_2 \dots b_n\}$.

The $M_R = [m_{ij}]_{m \times n}$ — matrix or relation R

where:

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_i) \in R \\ 0 & \text{if } (a_i, b_i) \notin R \end{cases}$$

Put 1 in matrix if ordered pair ∈ R and zero if not.

☐ Properties (Relation on Sets):

Reflexive: R is reflexive iff $m_{ii} = 1$ for $i = 1, 2 \dots n$.

* R is reflexive if all elements on the main diagonal of M_R are equal to 1.

Symmetric: R is symmetric iff $m_{ij} = m_{ji}$ for $i = 1, 2 \dots n$ & $j = 1, 2 \dots m$

folds diagonally into triangle & see values in mirror

Anti Symmetric: R is anti-symmetric if $m_{ij} = 1$ with $i \neq j$ then $m_{ji} = 0$.

Also if $m_{ij} = 0$ then $m_{ji} = 0$, when $i \neq j$.

Transitive couldn't be judge with simply watching a Matrix.

→ if mirrored elements (along diagonal) are either opposite or both zeros, then true else in case of both 1s, it's false.

Q: $R = \{(1,1),(1,2),(1,3),(1,4),(2,2),(2,4),(3,3),(4,4)\}$.

$$R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \longrightarrow (3,4) \notin R$$

- Reflexive as diagnol = 1.
- Symmetric (NOT) as $m_{41} \neq m_{14}$.
- Anti symmetric as all the mirrored elements are either opposite of each other or (0:0).

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Wednesday 19/11/25.

Ex: 9.1.5

Q: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$. The relation $R_1 = \{(1,1),(2,2),(3,3)\}$; $R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$. can be combined to obtain :

$R_1 \cup R_2 = \{(1,1),(1,2),(1,3),(2,2),(3,3)\}$ join

$R_1 \cap R_2 = \{(1,1)\}$ meet

$R_1 \oplus R_2$ = Union - Intersection $= \{(1,2),(1,3),(2,2),(3,3)\}$

$R_1 - R_2 = \{(2,2),(3,3)\}$

$R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$

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$\Longrightarrow$ <u>Composition:</u> R be relation from A to B
S be relation from B to C.

Then composition SOR is from A to C.

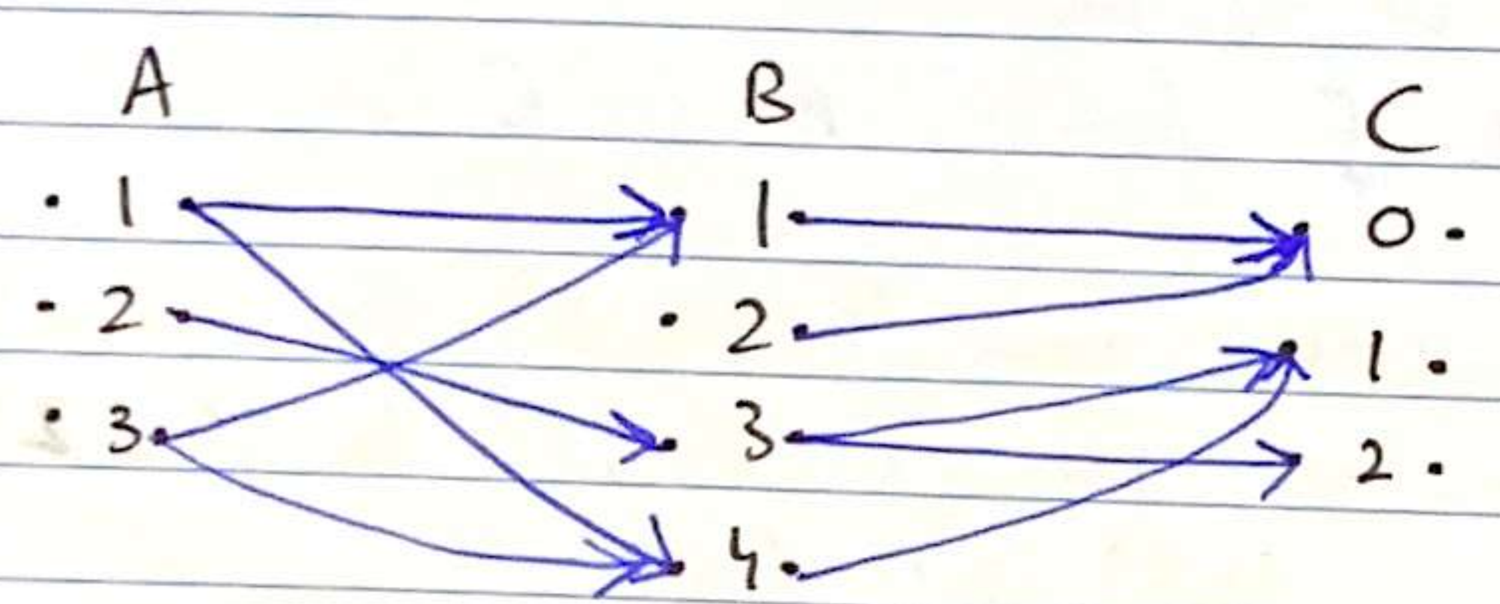
Ex: Let R be a relation from $\{1,2,3\}$ (A) to $\{1,2,3,4\}$ (B)
and S be a relation from $\{1,2,3,4\}$ to $\{0,1,2\}$ (C)
where $R = \{(1,1),(1,4),(2,3),(3,1),(3,4)\}$
$S = \{(1,0),(2,0),(3,1),(3,2),(4,1)\}$.
What is Composition SOR?

★ Take (1,1) from R and take 2nd element of it (i.e 1)
See where the first element of ordered pair is that 1.
(1,0) from S.
∴ answer will be (first of R, second of S)
partial ans : (1 , 0)

★ Now taking (1,4) from R.
S has (4,1) starting from 4. ∴ partial ans: (1,1).

$SoR = \{(1,0),(1,1),(2,1),(2,2),(3,0),(3,1)\}$.

SOR Graphically: Make arrows from both set of pairs of R & S



* Now take tail of first & move till head of last arrow. like seq: $1 \to 1 \to 0$ will form orderd pair $(1, 0)$

$2 \to 3 \to 1 = (2, 1)$ .....

☐ Representing Relations in Matrix:

$R_1 \cup R_2$ is 'join' of two zero-one matrices

$R_1 \cap R_2$ is 'meet' of two zero-one matrices

Def: We define:

$$M_{R_1 \cup R_2} = M_{R_1} \vee M_{R_2}$$

$$M_{R_1 \cap R_2} = M_{R_1} \wedge M_{R_2}$$

Q: Find join & meet of the zero-one matrices.

$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$

NOTE: order of A & B shld be same.

$A \vee B$ (join) $= \begin{bmatrix} 1 \vee 0 & 0 \vee 1 & 1 \vee 0 \\ 0 \vee 1 & 1 \vee 1 & 0 \vee 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$A \wedge B$ (meet) $= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$.

$\boxed{\text{Ex: } 9.3}$ Continue....

☐ Combining Relations:

⇒ Boolean Product:

Ex: Find the boolean prod. of zero-one Matrix

$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

NOTE: Both matrices shld be opposite order like 3x2 and 2x3.

Method: Simple Matrix multiplication

Bool prod/ Composition

$$A \odot B = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

- [ ] Transitive : A relation R defined on a matrix M_R is transitive if M_R is equivalent to its boolean square. i.e : $M_R = M_R^2$.

Q1. $M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

what are matrix rep. $R_1 \cup R_2$ & $R_1 \cap R_2$.
Do 'and' & 'or' operation entry by entry for $\cap$ & $\cup$ respectively.

Ex: 9.4 "Closure of Relations:"
A closure is a set of elements that are closed under some property P. → reflective, symmetric, transitive

Def: If R is a relation on set A, then the closure of R w.r.t P, if it exists, is the relation S on A with property P that contains R and is subset of A x A.

- The closure of a relation R wrt property P is the relation obtained by adding the minimum no. of ordered pairs to R to obtain property P.

Reflexive Closure: $S = R \cup \Delta$ → property P
where $\Delta = \{(a,a) \mid a \in A\}$
S is obtained by adding missing mirror images of $a \in A$.

Symmetric Closure: $S = R \cup R^{-1}$ → property P
where $R^{-1} = \{(b,a) \mid \dots\}$.
S is obtained by taking missing parallel symmetries missing in R.

Q: Let $R = \{(0,1),(1,1),(1,2),(2,0),(2,2),(3,0)\}$. defined on set $\{0,1,2,3\}$.
Find (a) Reflexive Closure
(b) Symmetric Closure.

(a) Δ = missing mirror pairs in R.
$\Delta = \{(3,3),(4,4)\}$
Closure $= R \cup \Delta$
$= \{(0,0),(1,1),(1,2),(2,0),(2,2),(3,0),(3,3)\}$

(b) (a,b) & (b,a) should be there.
R^{-1} = missing symmetric pairs
$R^{-1} = \{(1,0),(2,1),(0,2),(0,3)\}$
↳ since (1,0) isn't there for (0,1)

Symmetric $= R \cup R^{-1} = \{(0,1),(1,0),(1,1),(1,2),(2,1),(2,0),(0,2),(3,0),(0,3)\}$.

LECTURE-25

Monday 24/11/25.

Ex: 9.4 : Transitive Closure :

Theorum 2: The transitive closure of a relation R equal ~~the~~ connectivity relation R^* where $R^* = \bigcup_{n=1}^{\infty} R^n$

$= R^1 \vee R^2 \vee \dots$ → or sign

Theorum 3: Let M_R be the zero-one matrix of the relation R on a set of n-elements. Then the zero-one matrix of the transitive closure R^* is

transitive closure

$M_R^* = M_R \vee M_R^{[2]} \vee \dots \vee M_R^{[n]}$

GIST: Include the missing elements in set to complete transitive properties. E.g: if 'S' has (1,2) & does not have (2,1) ∴ we'll add (2,1) in set.

Q: Find the zero-one matrix of transitive closure of the relation R where:

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}_{3\times3}$$

If relation in set given, we'll form a matrix.

since size of matrix is 3x3 ∴ go uptill power of 3.

$M_R^* = M_R \vee M_R^{[2]} \vee M_R^{[3]}$

$$M_R^{[2]} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Trick for multiplication: if a (row x column), for a cell have 1 x 1 in any of (_x_) + (_x_) + (_x_) blank; the answer of that cell is directly 1; else 0.

$$M_R^{[3]} = M_R * M_R * M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$ → could be diff. than $M_R^{[2]}$.

$$M_R^* = \underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}}_{(M_R)^1} \vee \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{(M_R)^2} \vee \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{(M_R)^3}$$

$$M_R^* = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}.$$

∴ $M_R^* = \{(1,1), (1,2), (1,3), (2,2), (3,1), (3,2), (3,3)\}$.

Ex: 9.5 "EQUIVELENCE RELATIONS":

Def: A relation on a set **A** is called an equivalence relation if it is a reflexive, symmetric, transitive.

Def: Two elements a & b that are related by equivalence relation are called equivalent.

Notation:

$a \sim b$, $a R b$ → a is related to 'b'.

Q: Let R be the relation on the set of real no's s.t aRb iff a-b is an integer. Is R an equivalence relation.

$R = \{ aRb \mid a-b \in \mathbb{Z} \}$

Reflexive: aRa → always replace 'b' with 'a' in any ~~relation b/w 'a' & 'b'~~ for reflexive

$a - a = 0 \in \mathbb{Z}$

$\therefore$ it's reflexive

Symmetric: if aRb , $a-b \in \mathbb{Z}$

Let $a-b = c$ where $c \in \mathbb{Z}$

$\therefore b-a = -c$ $\therefore -c \in \mathbb{Z}$

so bRa is symmetric.

Transitive: $((aRb) \wedge (bRc)) \rightarrow aRc$.

★ prove aRc which is $a-c \in \mathbb{Z}$

For that; add and subtract 'b'

$a-b+b-c \rightarrow (a-b)+(b-c)$

★ since both $(a-b) \in \mathbb{Z}$ & $b-c \in \mathbb{Z}$ $\therefore$ $a-c \in \mathbb{Z}$ and $[aRc]$ ✓

Q: Let m be an integer, $m > 1$. Show that the relation $R = \{(a,b) \mid a \equiv b \pmod m\}$ is equivalence relation on set of integers.

$a \equiv b \pmod m$ $=$ $m \mid a-b$ $= \dfrac{a-b}{m} \in \mathbb{Z}$.

Reflexive: Replace 'b' with a $\therefore m \mid a-a$.

$= m \mid 0$

$\therefore a \equiv a \pmod m$ & reflexive holds.

Symmetric:

$m \mid a-b \rightarrow a-b = mk$, $k \in \mathbb{Z}$

for symmetric $\therefore b-a = -mk$

and still $m \mid b-a$. $\therefore b \equiv a \pmod m$ so symmetric

Transitive: Let $a \equiv b \pmod m$, $b \equiv c \pmod m$

$m \mid a-b$, $m \mid b-c$.

$a-b = mk$ ① $k \in \mathbb{Z}$, $b-c = ml$ ② $l \in \mathbb{Z}$

$\therefore$ now add ① & ②.

$(a-b)+(b-c) = a-c$

$\therefore a-c = mk + ml \rightarrow a-c = m(k+l)$

since 'k' & 'l' both Z $\therefore k+l \in \mathbb{Z}$ $\therefore m \mid a-c$ and $a \equiv c \pmod m$ ✓.

Equivalence Classes:

Def: Let R be an equivalence relation on a set A. The set of all elements $a \in A$ is called eq. class of. of a.

Notation: $[a]_R$

$[a]_R = \{ s \mid (a, s) \in R \}$

Eg:
↳ What are the equilance classes of 0,1,2,3 for congruence mod 4?

⇒ **Partition**: A partition of a set S is a collection of disjoint non-empty subsets that have 'S' as their union:

1) $A_i \neq \Phi$ — No empty subsets
2) $A_i \cup A_j = \Phi$ — No overlapping
3) $\cup A_i = S$ — all element involved

S

A_0	A_1	A_2
1	5	3
7	2	9
4	8	6

when we can take union of the partitions formed (on any specific property), it will form the orignal set.
$A_0 \cup A_1 \cup A_2 = S$; so it's a partition.

★ Each partition of a set is an equilance class i.e: A_0, A_1, A_2.

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for Z (integers)

$A = \{0, 1, 2, 3\}$.

To show 'A' as equivalence class↑, we'll do:

0 (mod 4) : remainder 0
$A_0 = [0]_4 = \{ \ldots, -8, -4, 0, 4, 8 \ldots \}$.

1 (mod 4) : remainder 1
$A_1 = [1]_4 = \{ \ldots -7, -3, 1, 5, 9 \ldots \}$.

2 (mod 4) : remainder 2
$A_2 = [2]_4 = \{ \ldots -2, 2, 6, 10 \ldots \}$.

3 (mod 4) : remainder 3
$A_3 = [3]_4 = \{ \ldots$

LECTURE - 26

Wednesday.
26/11/25.

Ex: 10.1 :

GRAPHS

Def: A graph $G = (V, E)$ consists of V, a nonempty set of vertices and E, a set of edges. Each has either one or two vertices associated with it, called its end points. An edge is said to connect its endpoint.

~~Undirected~~ Infinite Graph: has infinite no. of vertices & edges
Finite Graph: has finite " " " " "

Undirected Graphs:

⇒ Simple Graphs: A graph in which each edge connects two diff vertices and where no two edges connect same pairs of vertices.

→ Best Definations from Book.

⇒ Multigraphs: A graph in which multiple edges connect s the same pair of vertices. THEY SHLD have same Direc[redacted]

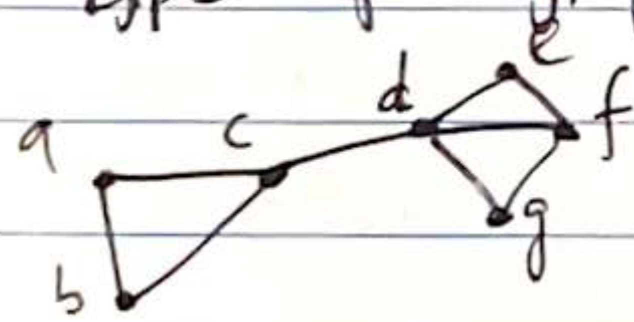
⇒ Psuedo graph: A graph may include loops, and possibly multiple edges containing same pairs of vertices or a vertex to itself are sometimes called psuedograph.

Types	Directed / Undirected Graph	Multiple edges allowed	Loops Allowed
Simple	Undirected	No	No
Multigraphs	Undirected	yes	No
Psuedo graphs	Undirected	yes/No	yes
Simple directed	directed	No	No
Directed Multigr.	directed	yes	Yes/No.
Mixed graphs	directed/Und.	yes	Yes.

edges with some vertices are directed & with some undirected

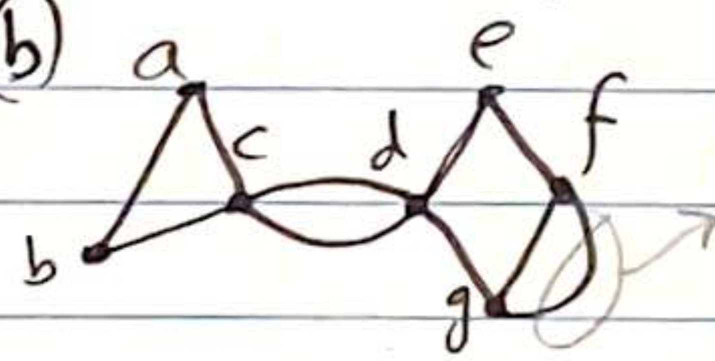
Q: Type of graphs. Directed/Undirected with further types.

(a)



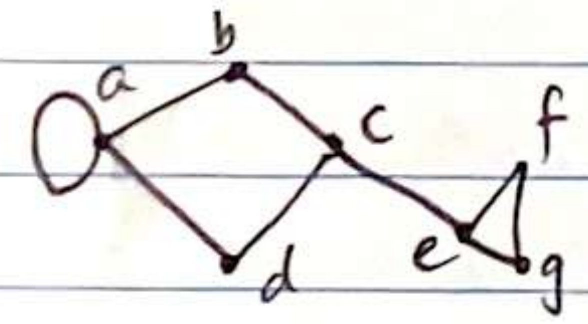
Simple Undirected

(b)



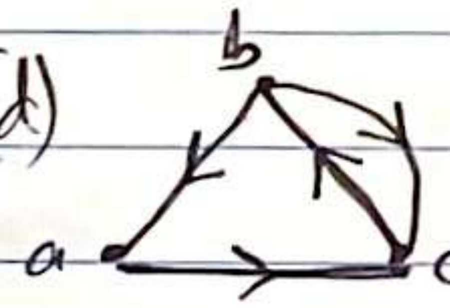
Undirected Multi-graph

(c)



Puesdo-Undirected

(d)

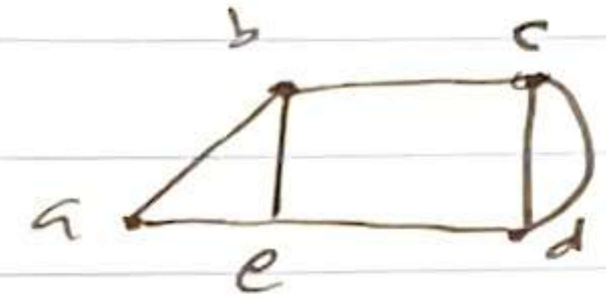


Simple directed

as two arrows from b→c & c→b have diff direction.

Q:

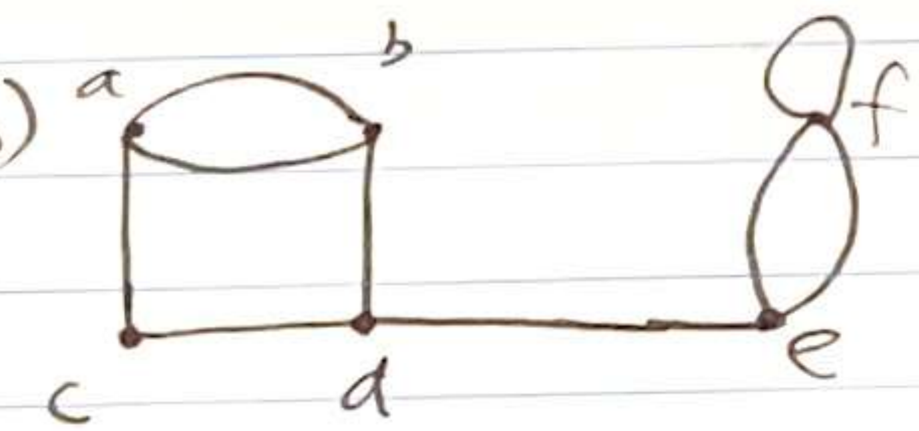
(a)



Undirected Multi.

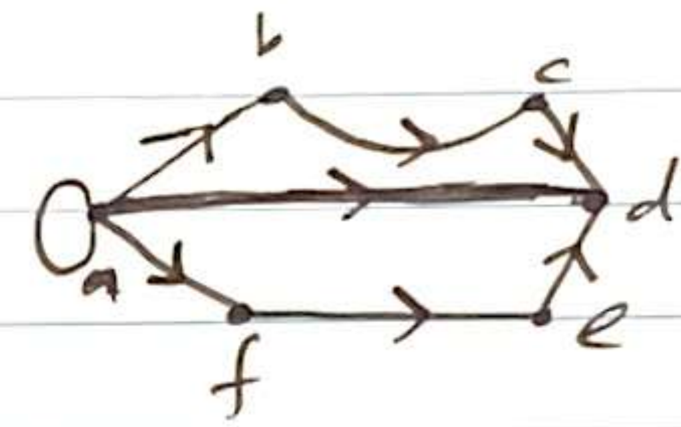
(Since no direction so both ways)

(b)



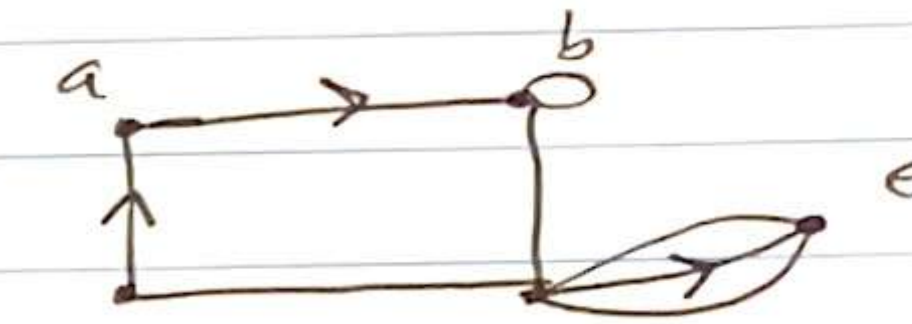
Undirected Psuedo.

(c)



directed Multi-graph.

(d)



Directed Mixed.

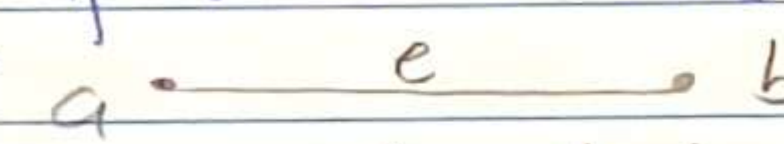
(e)

Undirected Simple

Ex: 10.2 "Graph Terminology & Special Types of Graph".

Def:

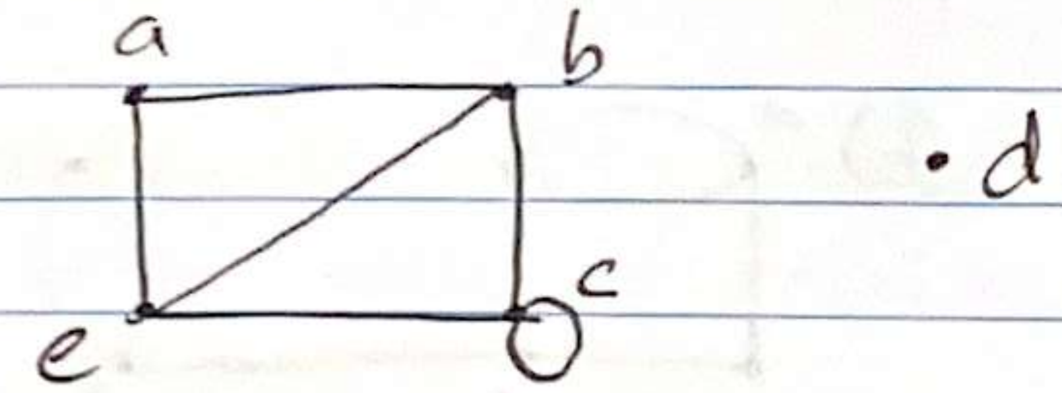
Two vertices 'u' and 'v' in an undirected graph 'G' are called adjacent in G if u & v are endpoints of a same edge 'e'



e is incident on vertices a & b. a adjacent to b.

Def: The set of all neighbours of a vertice 'v' of $G = (V, E)$, denoted by $N(v)$ is called neighbourhood of 'v'

Q:



N(a) = { b, e }	deg (a) = 2
N(b) = { a, c, e }	deg (b) = 3
N(c) = { c, b, e }	deg (c) = 4
N(d) = { }	deg (d) = 0
N(e) = { a, b, c }	deg (e) = 3

⇒ Degree: The degree of a vertex of an undirected graph is the no. of edges incident with it, except that a loop at vertex contributes twice the degree of that vertex.

Notation: deg (v).

⇒ Isolated: Vertex of degree zero (e.g: d in above)

⇒ Pendant: Vertex of degree one.

$\longleftrightarrow$

LECTURE - 27 | Monday. 1/12/25.

Ex: 10.2 "Terminologies of Graphs:

Def: Directed Graphs: Let $u \longrightarrow v$

v is said to be adjacent from u.

u is said to be adjacent to ~~[illegible]~~ v.

u: initial vertex

v: terminal vertex.

loop has same initial and terminal state.

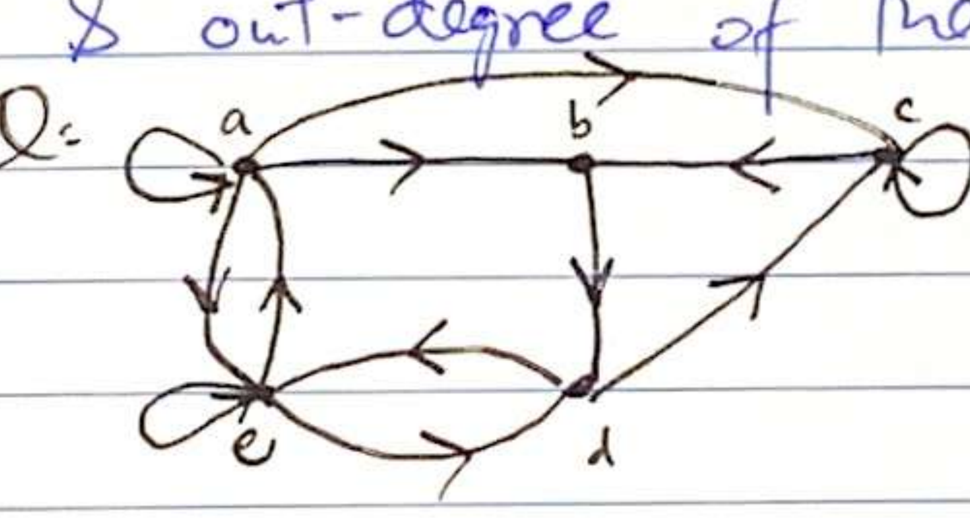
Def: In a graph with directed edges the in-degree of a vertex 'v' is denoted as $deg^-(v)$, is the no. of edges with 'v' as their terminal vertex.

The out-degree of v, denoted as $deg^+(v)$, is the no. of edges with v as the ~~terminal~~ initial vertex.

Note:

Loop at a vertex contributes 1 to both the in-degree & out-degree of the vertex.

Q:



Find degree of each vertex:

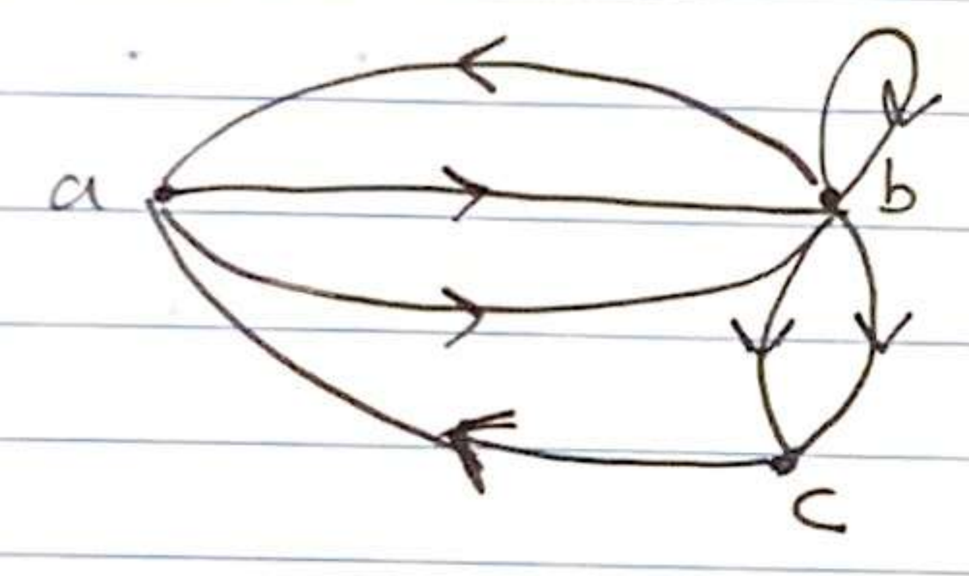
Degree-in	degree out
a = 2	a = 4
b = 2	b = 1
c = 3	c = 2
d = 2	d = 2
e = 3	e = 3

⇒ Theorum 3: Let $G(V, E)$ be a directed graph.

Then: Sum of $deg^-(a)$ = Sum of $deg^+(a)$ = no. of edges

$$\sum_{a \in V} deg^-(a) = \sum_{a \in V} deg^+(a) = |E| = 12$$

Q:



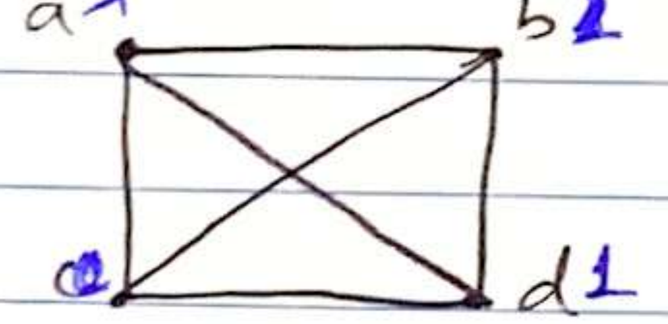
	+ (out)	− (in)
a	2	2
b	4	3
c	1	2
d	1	1

⇒ Theorem 1: Handshaking Theorum (Undirected)

Let $G = (E, V)$ be undirected graph with 'm' edges. Then:

$$2m = \sum_{a \in V} deg(a).$$

Q:



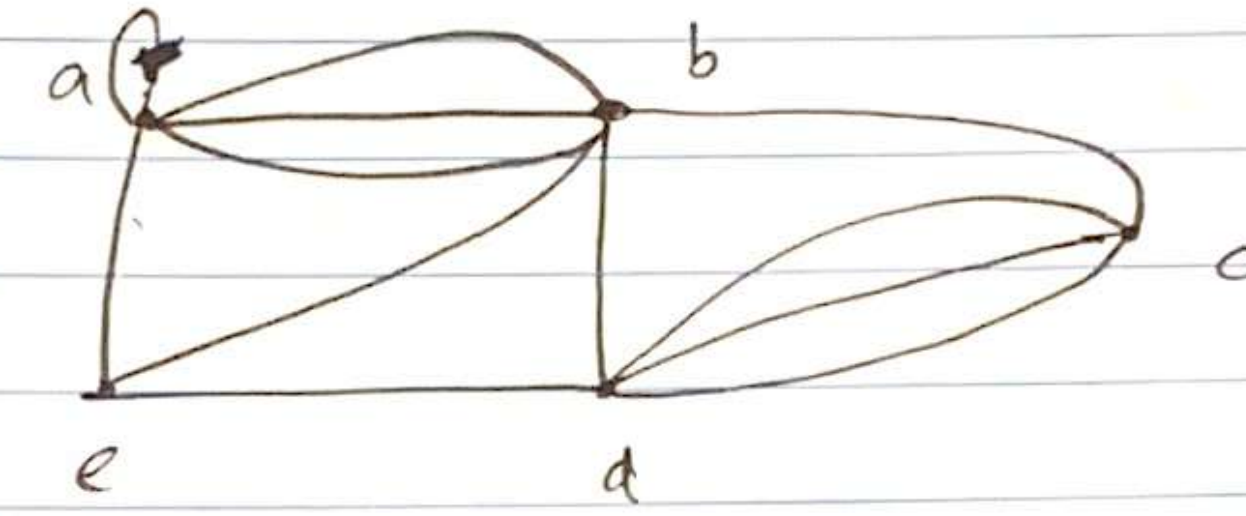
degree of each node/vertex = 4.

$\therefore \sum_{a \in V} deg(a)$

$3 + 2 + 1 + 0 = \boxed{8}$

$= 4 + 4 + 4 + 4 = 16$

$2m = 16 \rightarrow \boxed{m = 8}$ ← edges.



(a) Find no. of vertex & edges → $V = 5$, $E = 12$

(b) Each vertex's degree.

(c) prove theorum 1

(b) $a = 6$ $d = 5$

$b = 6$ $e = 3$

$c = 4$

(c) $2m$ = Sum of degrees

$2m = 6 + 6 + 4 + 5 + 3$

$m = 24/2$ → $\boxed{m = 12}$

proved. (same edges)

⇒ Def: TYPES OF SIMPLE GRAPHS:

A simple graph G is called bipartite if its vertex set V can be partioned into two distinct sets V_1 and V_2 such that every edge in the graph connect a vertex in V_1 and vertex in V_2 (so that no edge in G connects either any two vertex in V_1 or any two vertices of in V_2).

We call pair (V_1, V_2) a bipartition

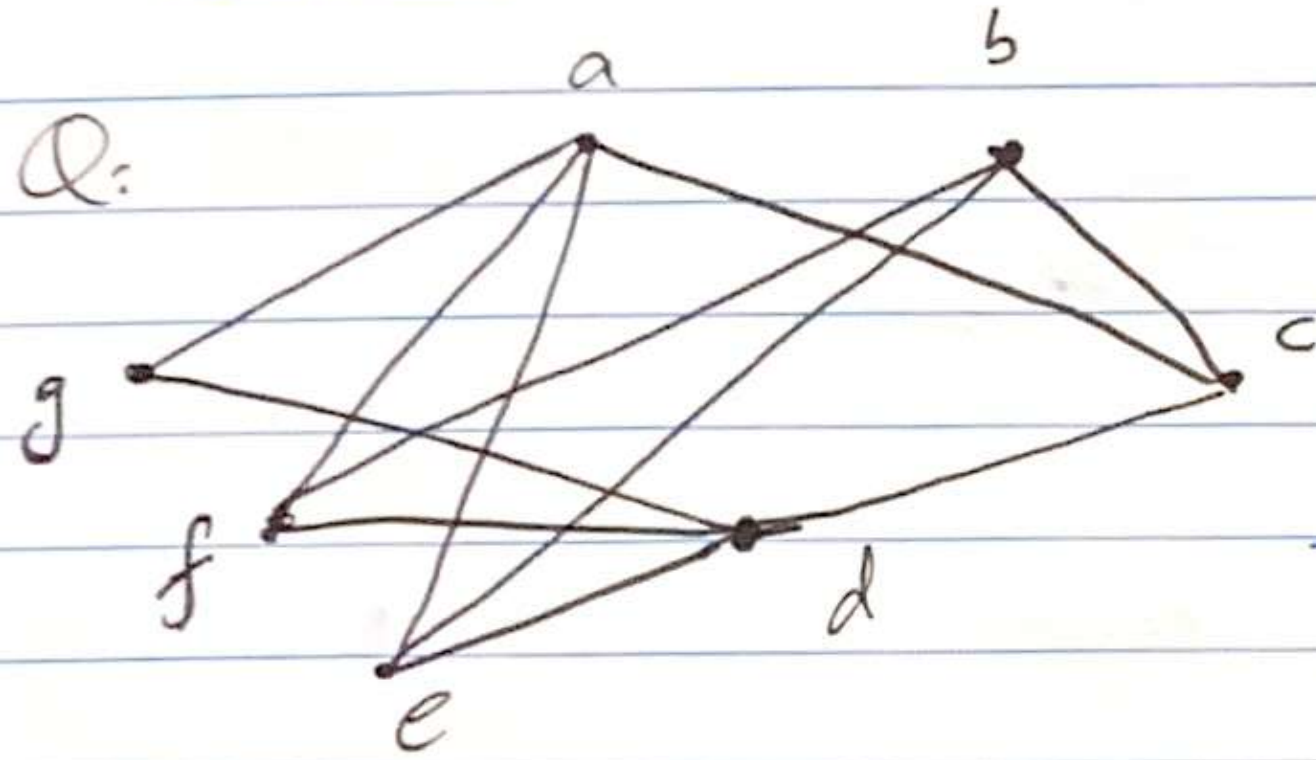
$= \{a, b, c, d, e\}$

$V_1 = \{a, c, d\}$ → a, c, d does not have any connection b/w each other

$V_2 = \{b, e\}$ → b, e doesn't have connection with e/o

a, c, d & b, e have connections together

Q: Is the graph Bipartite



V_1	V_2	
a	c, e, f, g	neighbours of 'a'
b	c, e, f	neighbours of 'b'
d	c, g, f, e	neighbours of 'd'

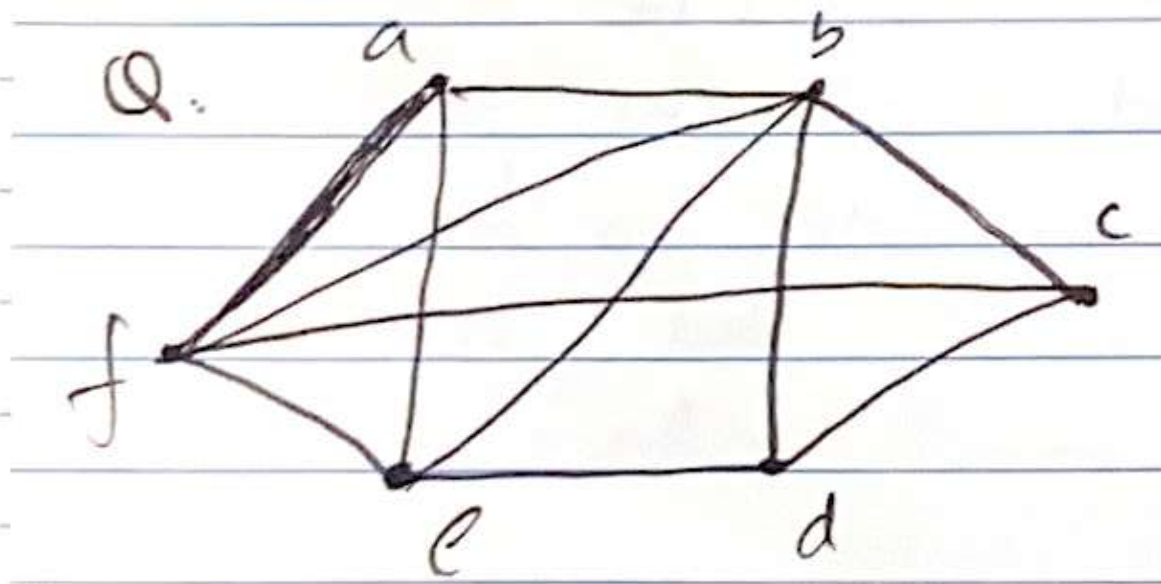
skipped 'c' since it's in V_2

$V_1 = \{a, b, d\}$

$V_2 = \{c, e, f, g\}$.

★ we also have to check if the neighbours of each element in V_1 i.e: c, e, f, g (from $V_1 = a$) should not be neighbours of e/o else graph is "non bipartite".

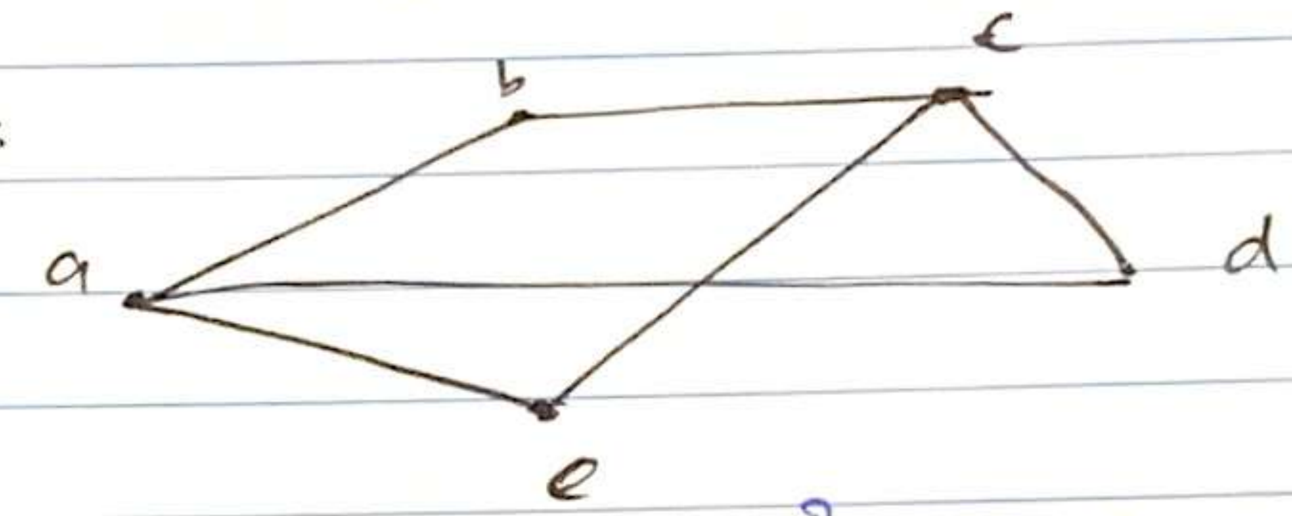
Q.



V_1	V_2
a	b, e, f
b	a, c, d, e, f
c	b, d, f
d	b, c, e
e	a, b, d, f
f	a, b, c, e.

★ Since 'b and e' have common edge and they both are in 'V_2' ∴ NOT BIPARTITE.

Q:



V_1	V_2
a	b, d, e
c	b, d, e

$V_1 = \{a, c\}$

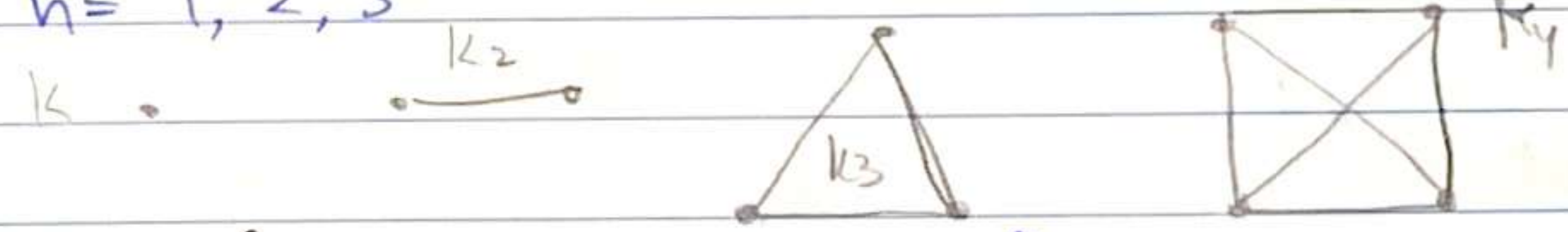
$V_2 = \{b, d, e\}$

Perfectly Bipartite since same and equal no. of V_2 elements for each V_1 element.

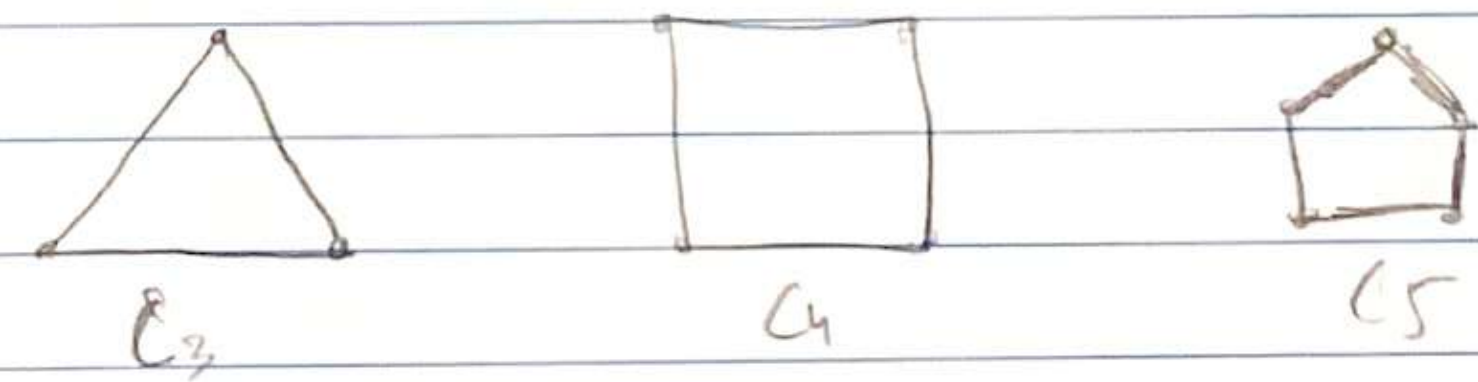
Complete Graph:

A complete graph on 'n' vertices, is denoted by K_n is a simple graph that contains exactly one edge b/w each pair of distinct vertices

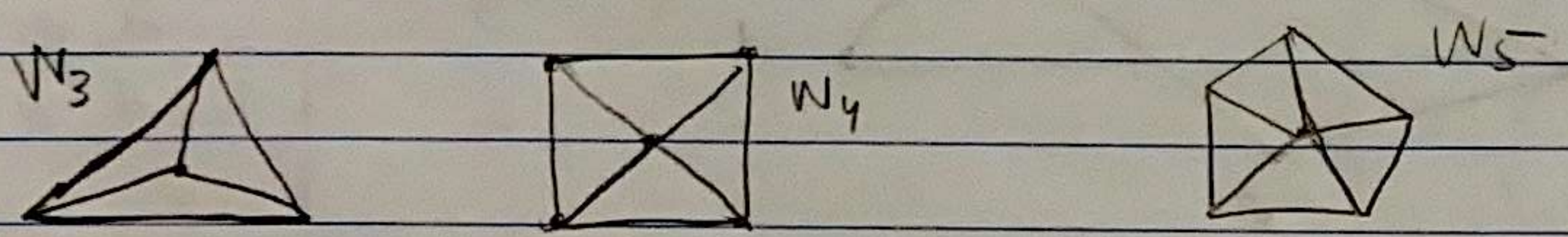
$n = 1, 2, 3 \ldots$



⇒ Cycles: A circle C_n, $n \geq 3$ consists of n vertices $v_1, v_2 \ldots v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}$ and $\{v_n, v_1\}$



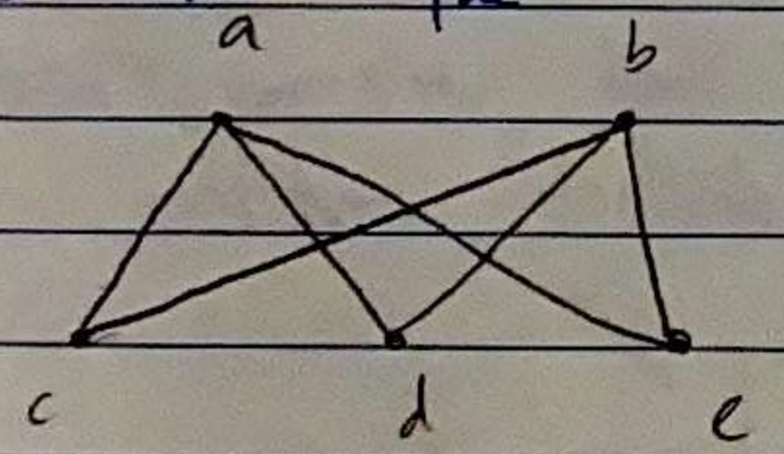
⇒ **Wheels**: We obtain a wheel W_n when we add an additional vertex to a cycle C_n, for $n \geq 3$ and connect this new vertex to each of 'n' vertix.



☐ Complete/ perfect Bipartite Graph:

K_4 : Complete graph
$K_{1,4}$: Complete Bipartite

A complete bipartite graph $K_{m,n}$ is a graph that has its vertex set partitioned into two disjoint subset of m & n vertices iff one vertix is in first subset and other vertex is in the second subset.

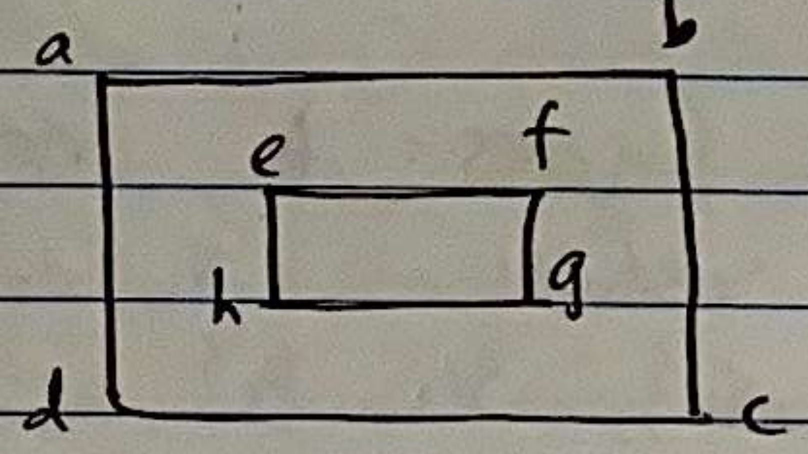


$K_{2,3}$

V_1	V_2
a	c, d, e
b	c, d, e

Q: Bipartite or Not?

V_1	V_2
a	b, c, d

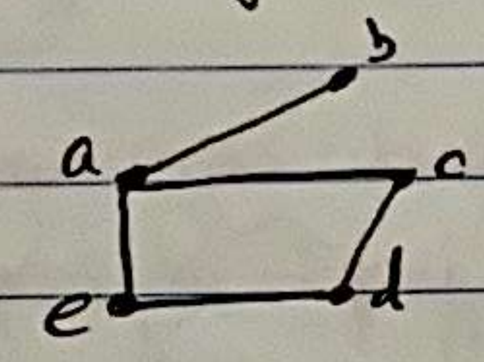


LECTURE-28

Wednesday. 3/12/25.

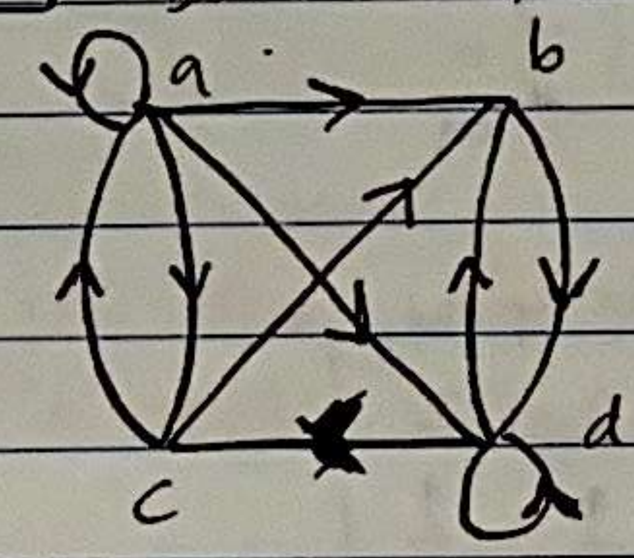
Ex: 10.3 "Representing Graphs".

☐ Adjacency List of Simple Graph:



Vertex	Adjacent vertex
a	b, c, e
b	a
c	a, d
d	c, e
e	a, d

☐ Directed GRAPHS:



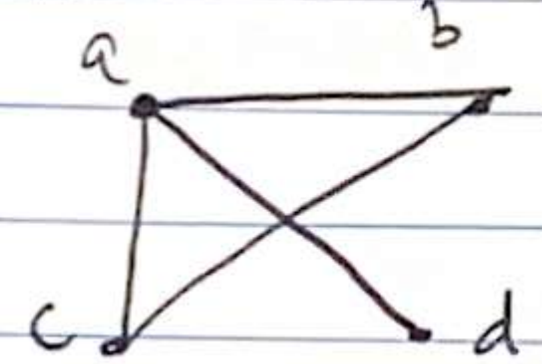
Initial/outward arrow

Initial Vertex	Terminal vertex
a	a, b, c, d
b	d
c	a, b
d	b, c, d.

⇒ Adjacency Matrix: Suppose $G = (V, E)$ a simple graph with n vertix $v_1, v_2, v_3, \dots v_n$. The adjacency matrix is $n \times n$ matrix

$$A_G = [a_{ij}]_{n \times n}, \text{ where } a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

Ex: Use adj. matrix to represent a graph.



no. of vertices $= 4 \rightarrow n$

$\therefore$ 4x4 matrix.

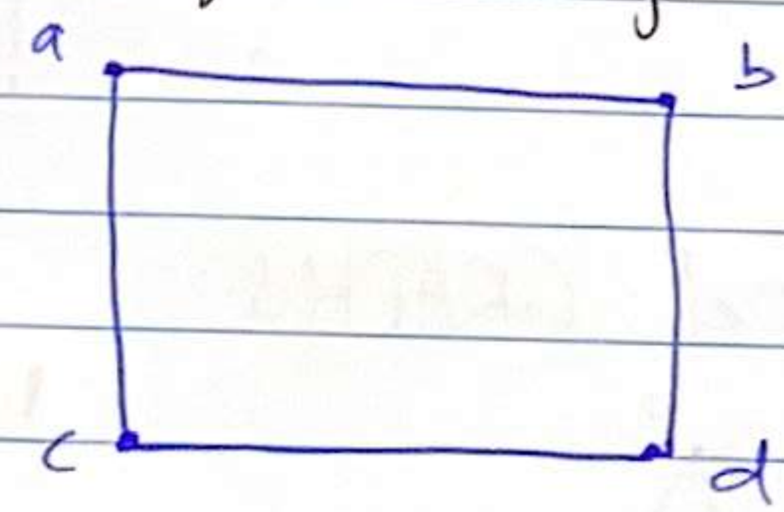
$$\begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 1 & 1 \\ b & 1 & 0 & 1 & 0 \\ c & 1 & 1 & 0 & 0 \\ d & 1 & 0 & 0 & 0 \end{array}$$

* Symmetric over diagonal.

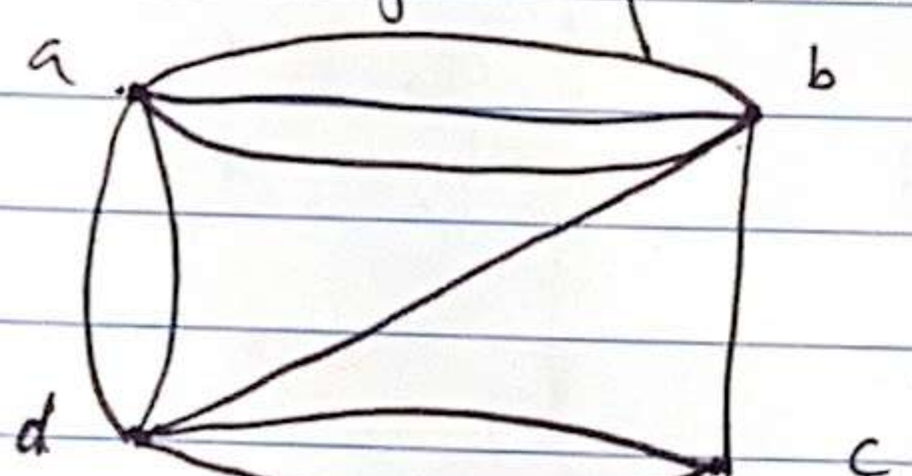
diagonal zero since no loops (in simple).

Ex: Draw graph with the given adjacent metrix

$$\begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 1 & 0 \\ b & 1 & 0 & 0 & 1 \\ c & 1 & 0 & 0 & 1 \\ d & 0 & 1 & 1 & 0 \end{array}_{4\times 4}$$



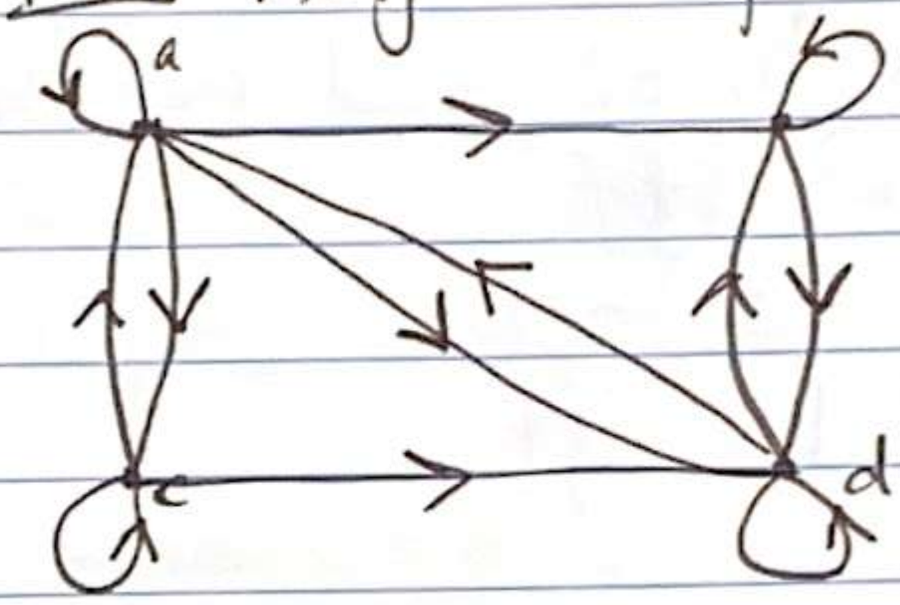
☐ Adjacency Matrix of Psuedo Graph



$$\begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 3 & 0 & 2 \\ b & 3 & 0 & 1 & 1 \\ c & 0 & 1 & 1 & 2 \\ d & 2 & 1 & 2 & 0 \end{array}_{4\times 4}$$

no. of edges b/w 'a' & 'd'

☐ Adj. of Directed Graphs:

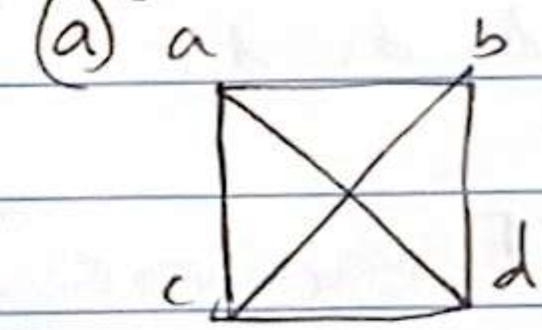


$$\begin{array}{c|cccc} & a & b & c & d \\ \hline a & & & & \\ b & & & & \\ c & & & & \\ d & & & & \end{array}_{4\times 4}$$

Q: Draw Graph & Adj. Matrix of:

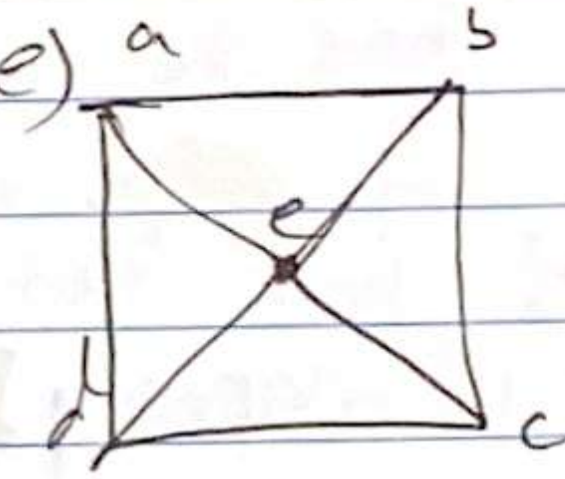
(a) K_4 (b) $K_{1,4}$ (c) $K_{2,3}$ (d) C_4 (e) W_4.

(a)



	a	b	c	d
a	0	1	1	1
b	1	0	1	1
c	1	1	0	1
d	1	1	1	0

(e)



	a	b	c	d	e
a	0	1	0	1	1
b	1	0	1	0	1
c	0	1	0	1	1
d	1	0	1	0	1
e	1	1	1	1	0

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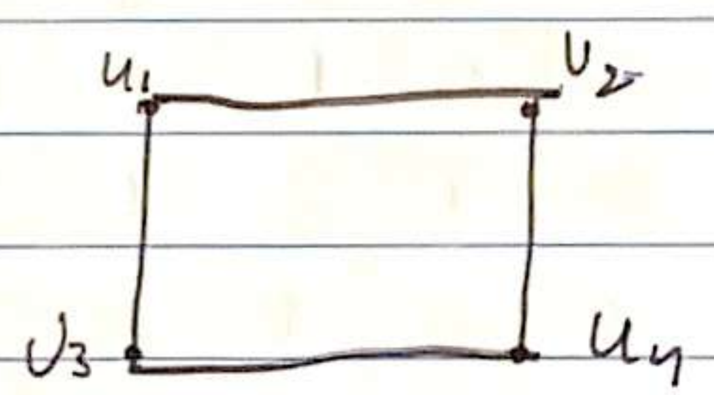
10.3: Isomorphism of Graph.

Def: The simple graph $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic (i.e: $G_1 \cong G_2$) iff
(i) f is bijective (one-to-one & onto)
(ii) $\forall\, a, b \in V_1$, $\{a, b\} \in E_1$ iff $\{f(a), f(b)\} \in E_2$.

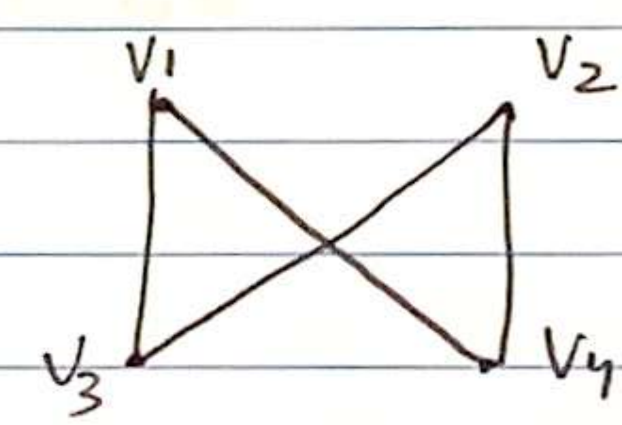
0, 2, 3, 1 — three vertices of degree = 2

NOTES: We say graph G and H are isomorphic if.
(a) G & H have the same no. of edges & vertices
(b) G & H have the same degree seq. $D = d_0, d_1, d_2 \ldots$
(d_i = no. of ~~[illegible]~~ vertices of degree i),
(c) G & H have same Structure Cycle (OF each vertex)
(d) Show (1-1) mapping

Q: G



H



1) Vertices → $G = H = 4$
Edges → $G = H = 4$

2) Degree Seq. — 4 vertices with degree = 2

$D_G = \underset{d_0}{0},\ \underset{d_1}{0},\ \underset{d_2}{4}$
$D_H = 0, 0, 4$

(sum of it should be equal to no. of vertices
$\therefore 0 + 0 + 4 = 4$).

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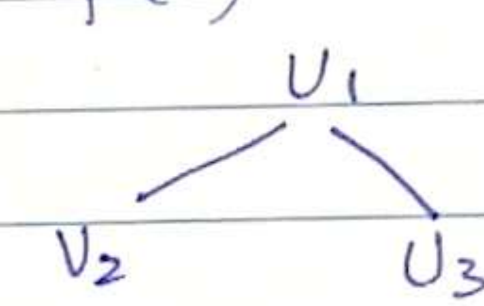
- But always choose the vertex that has degree = 1 (or minimum).
- can start with any vertex of any degree.

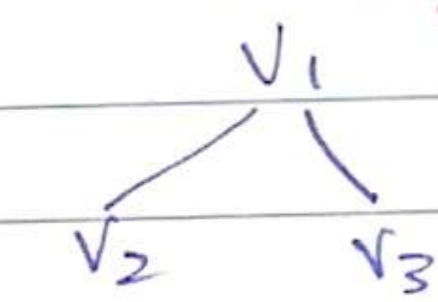
3) we gonna map u_1 to it's image in H.

(For the image of u_1; we need to see its degree and the vertices deg. adjacent to u_1).

$\deg(u_1) = 2$
u_1 is adj to u_2 & u_3 — of both u_2 & u_3 $\therefore$ adj = **2**

Set $f(u_1) = v_1$ → we took v_1 since its properties i.e: degree and degree of it's adjacent vertex are equal.

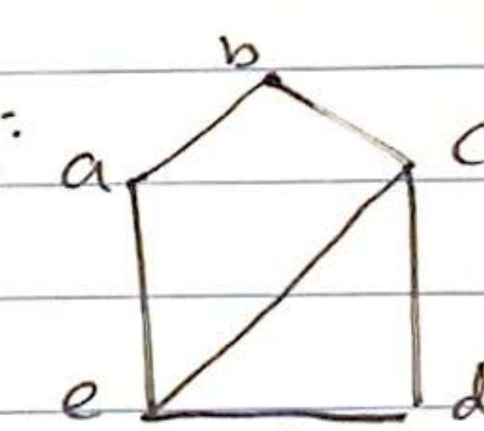




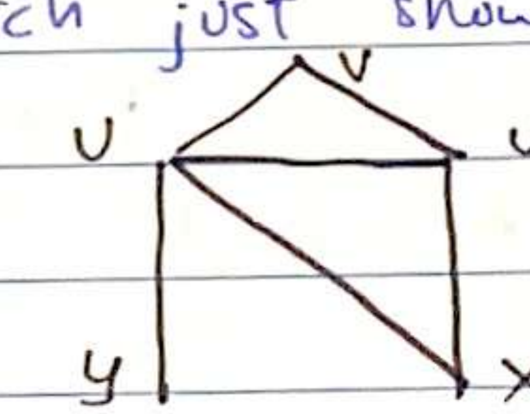
$f(u_2) = v_3$
$f(u_3) = v_4$
$f(u_4) = v_2$
} Map by own desire. This is 1-1 mapping which just shows isomorphism is possible

$\therefore G \cong H$
(G is isomorphic to H)

Q: G



H

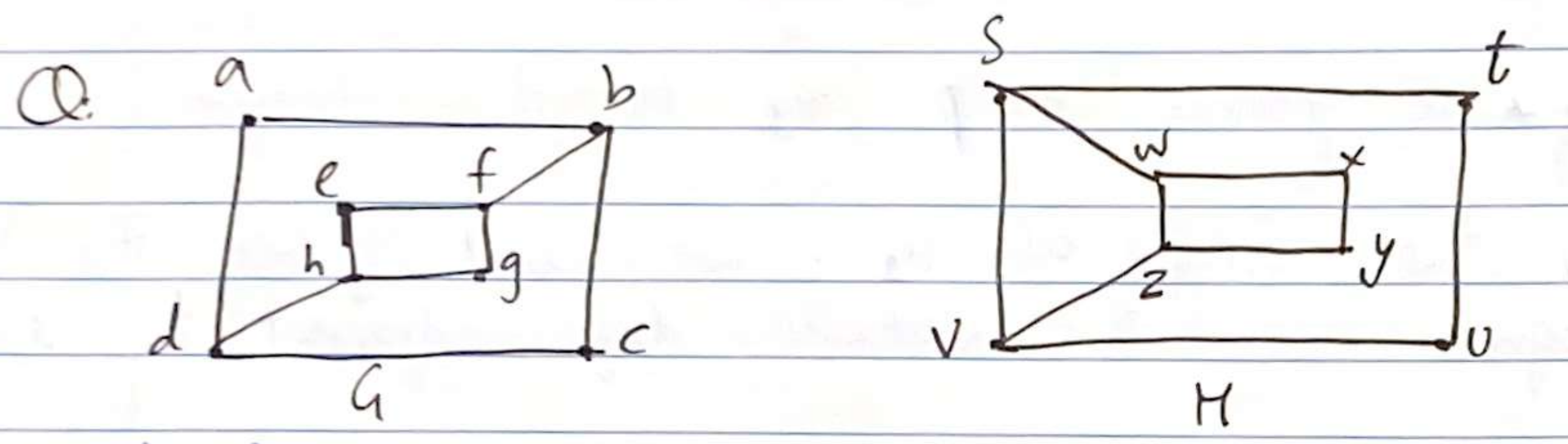


1) Edges ⇒ 6 = 6
vertices ⇒ 5 = 5

2) $D_G = 0, 0, 3, 2$
$D_H = 0, 1, 2, 2$

Since graph H has a vertex of deg **1** but G does not have any vertix of degree **1**.
$\therefore G \neq H$.

Q.



1) V = 8
E = 10

2) D_G = 0, 0, 4, 4
D_H = 0,0 , 4, 4

3) Image making for 'a'.
deg (a) = 2
deg of all adjacent vertices of 'a' = 3 (i.e: of b, d)
Whereas, we can map 'a' only to x, t, u, y
as Only these four vertices in H have degree = 2.

BUT all vertices t, u, x, y have adjacent vertices with degree 2 and 3 (which isn't 3 & 3).
$\therefore$ $G \ncong H$.

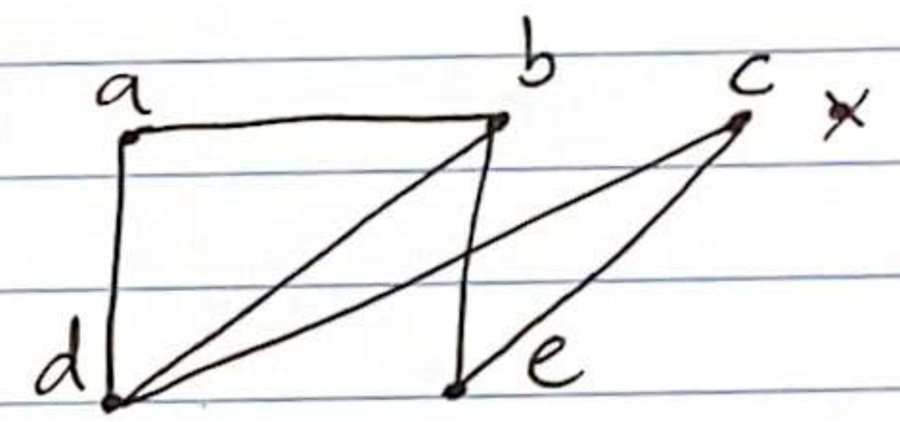
Monday
8/12/25.

LECTURE-29

⇒ Practice Q/s of Graphs. Picture in phone.

Q1: Represent adjacency List & adjacency matrix of given graphs.

(a)



vertex	adjacent vertex
a	b, d
b	d, e
c	d, e
d	a
e	b, c

Matrix

$$\begin{array}{c|c} & \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \\ e \end{array} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$